FoodShare – Implementation Plan

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Approach and phases

The plan delivers a working pilot in 8 weeks, starting Monday Oct 26, 2026, by building the backend core first and layering real-time, maps and analytics on top.

Assumptions: one to two full-stack developers and one part-time designer; a single pilot city; a seed group of donors, one or two NGOs and about 10 volunteers recruited in parallel.

Phase	Weeks	Dates	Outcome
0. Setup and design	1	Oct 26 to Oct 30	Repo, CI, environments, design tokens, clickable screens
1. Backend core	2 to 3	Nov 2 to Nov 13	Auth, RBAC, listings, geo feed, status transitions
2. Client and real-time	4 to 5	Nov 16 to Nov 27	Role dashboards, map feed, Socket.io live updates
3. Analytics and polish	6	Nov 30 to Dec 4	Impact dashboard, notifications, empty and error states
4. Hardening	7	Dec 7 to Dec 11	Tests, security pass, performance check, staging sign-off
5. Pilot launch	8	Dec 14 to Dec 18	Production deploy, onboarding of pilot users, feedback loop

Tasks and deliverables by phase

Phase 0: Setup and design

- [ ] Create repository, lint and format rules, and CI that runs tests
- [ ] Provision MongoDB Atlas and staging hosting; store secrets
- [ ] Finalise design tokens and key screens from the UI/UX document

Phase 1: Backend core

- [ ] User and Listing models with the 2dsphere index
- [ ] Register, login, JWT middleware and role guard
- [ ] Create listing, nearby feed with `$near`, my listings, pickup board
- [ ] Atomic claim, pickup, deliver and cancel transitions
- [ ] API tests for every endpoint and role

Phase 2: Client and real-time

- [ ] React app with auth, role routing and API client
- [ ] Nearby feed with Leaflet map, radius slider and filters
- [ ] Post-surplus form with map pin picker
- [ ] Socket.io client: new, update and status events; reconnect refetch
- [ ] Status stepper and role action buttons

Phase 3: Analytics and polish

- [ ] Impact endpoint and dashboard with stat tiles and charts
- [ ] In-app notifications for claim and delivery
- [ ] Expiry handling job and empty, loading and error states

Phase 4: Hardening

- [ ] Rate limiting, input validation review, CORS lock-down
- [ ] Concurrency, geo and socket tests; load test of the nearby query
- [ ] Accessibility check against the UI/UX document

Phase 5: Pilot launch

- [ ] Production deploy with backups and uptime monitoring
- [ ] Onboard donors, NGOs and volunteers; short how-to guide
- [ ] Weekly review of success metrics from the PRD

Milestones

Date	Milestone	Gate to pass
Oct 30	Design and setup complete	Tokens and screens approved; CI green

Date	Milestone	Gate to pass
Nov 13	Backend API complete	All endpoint tests pass; claim race test passes
Nov 27	Live feed demo	Two browsers see a status change within 2 seconds
Dec 4	Feature complete	Dashboard, notifications and expiry done
Dec 11	Release candidate	No open high-severity bugs; security checklist done
Dec 18	Pilot live	Production deployed; first donor, receiver and volunteer onboarded

Testing and quality

- **Every pull request:** lint, unit and API tests must pass before merge.
- **Required tests:** concurrent claim returns one success and one 409; radius query excludes a listing at 6 km for a 5 km search; each role is blocked from the other roles' endpoints.
- **Manual test script:** one donor, one receiver and one volunteer complete a full delivery on three devices.
- **Exit criteria for release candidate:** no open high-severity bugs, nearby query under 300 ms on a seeded database of 100,000 listings, accessibility checklist passed.

Deployment and launch

1. Deploy to staging on every merge to main and run the manual test script.
2. Take a database backup, then promote the same build to production after approval.
3. Set production environment variables and confirm HTTPS and CORS.
4. Seed the pilot area with partner NGOs and volunteers before opening donor sign-ups.
5. Soft launch to a small donor group for the first week; widen after the success metrics in the PRD are tracked.

Rollback: redeploy the previous build; database changes are additive, so no data migration is needed to roll back.

Risks and mitigations

Risk	Impact	Mitigation
Too few volunteers, so claimed food waits	Food expires; receivers lose trust	Recruit volunteers through partner NGOs first; allow receivers to self-collect later
Food-safety or liability concerns	Legal exposure	Donor declaration, expiry limits, disclaimer; legal review before launch
Duplicate claims	Wrong delivery	Atomic status transitions and a concurrency test
Real-time drift after network loss	Stale feed	Refetch on reconnect; REST is the source of truth
Schedule slip	Late pilot	Cut notifications and ratings first; keep the core delivery flow
Spam or fake listings	Wasted trips	Rate limits; verify organisation donors